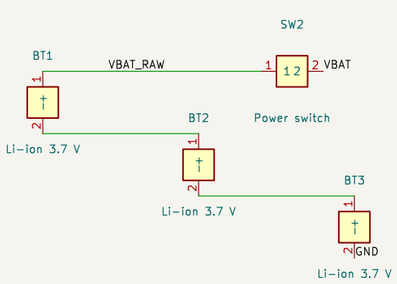
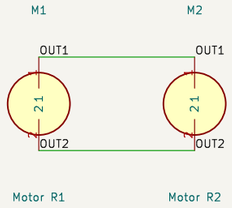
POWER (3 x Li-ion in series, about 11.1 V)

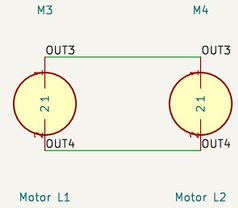


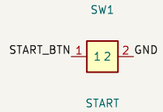


RIGHT SIDE: M1 || M2 (ENA/IN1/IN2)

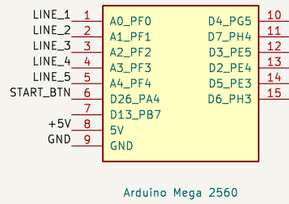


LEFT SIDE: M3 || M4 (ENB/IN3/IN4)

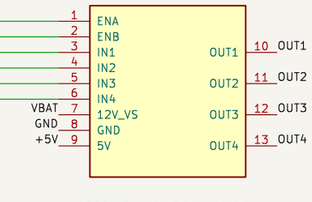




CONTROLLER (ARDUINO MEGA 2560)

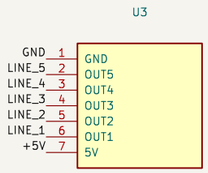


MOTOR DRIVER



D13 = on-board status LED (no external wiring)

LINE SENSOR (UGE electronics, 5 channels)



HARDWARE NOTES - Phase 0

1. Power: three 3.7 V Li-ion cells in series (BT1-BT3 = about 11.1 V, 12.6 V full) with a 3S protection board, through power switch SW2 to the L298N 12V (VS) terminal.
   C1 (1000 uF, 25 V or more) is fitted close to the L298N supply terminals.
2. The L298N 5V-EN jumper is fitted: its on-board regulator feeds the Mega 5V pin and the sensor 5V pin (+5V rail), so the Arduino has no separate supply.
   At 12.6 V (fresh from the charger) that regulator runs warm. If it gets hot, remove the jumper and feed the Mega 5V pin from a 5 V buck converter. Unplug the battery or the 5V link while the Mega is on USB.
3. Battery -, L298N GND, Mega GND and sensor GND are all common. ENA / ENB jumpers are REMOVED: wheel-speed PWM comes from Mega D4 (ENA, OC0B, Timer0) and D7 (ENB, OC4B, Timer4).
4. Direction pins: IN1 = D3, IN2 = D2 (right side, OUT1/OUT2); IN3 = D5, IN4 = D6 (left side, OUT3/OUT4). Right motors M1 || M2 on OUT1/OUT2, left motors M3 || M4 on OUT3/OUT4 (two motors in parallel per channel).
5. The motors are rated well below the battery voltage, so the firmware caps the PWM duty (MAX_SPEED in config.h). The start push-button (D26 to GND, INPUT_PULLUP) is optional: 's' over Serial also starts the run.
6. UGE 5-channel line sensor on analog pins: OUT1..OUT5 (left to right: L, CL, C, CR, R) on A0..A4. If the robot steers the wrong way set LINE_REVERSE_ORDER in config.h.
7. Only the Mega pins used in Phase 0 are drawn. Source of truth for pins: include/config.h (namespace cfg).

4 DC motors (2 per side, in parallel), 3 x Li-ion cells in series
Arduino Mega 2560 + L298N motor driver + 5-ch IR line sensor
ECNG 4504 - Embedded Systems for Wireless Communications - Group 6

Sheet: /
File: maze_robot.kicad_sch

Title: Maze Robot - Phase 0 Hardware Schematic

Size: A3	Date: 2026-10-15	**Rev: 1.3**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 1/1